

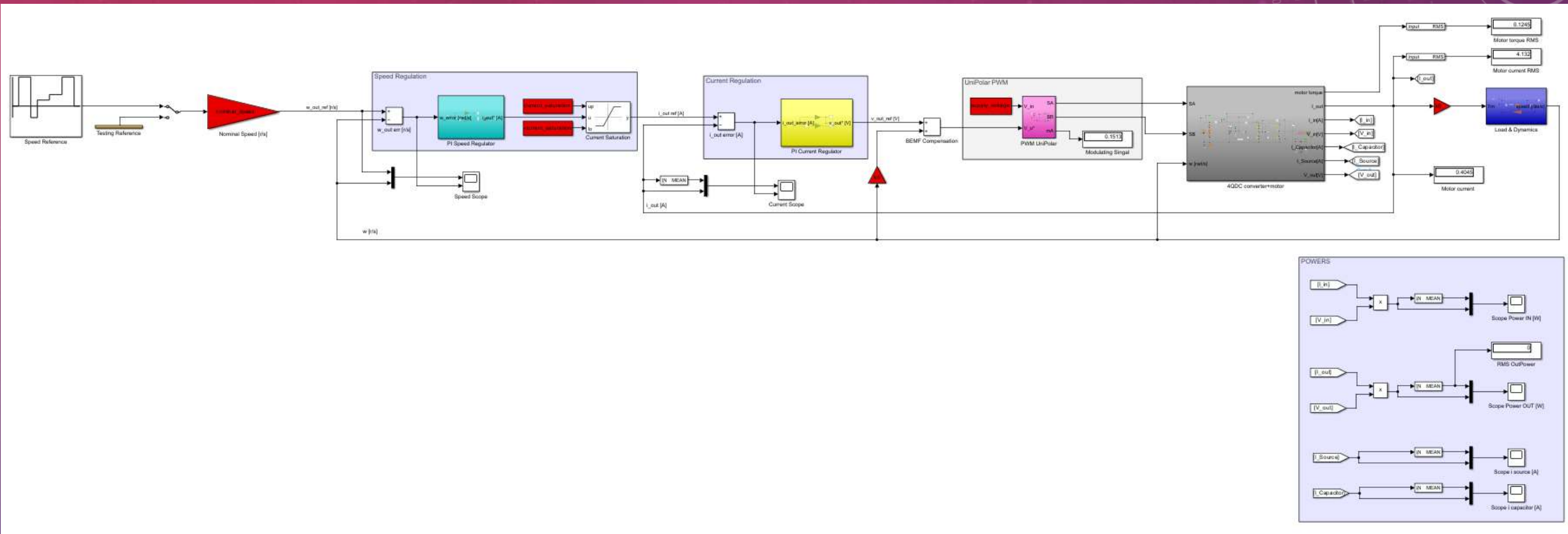
LABORATORY OF ELECTRIC DRIVES

Michael Quattrini – Final Report 14/02/2025



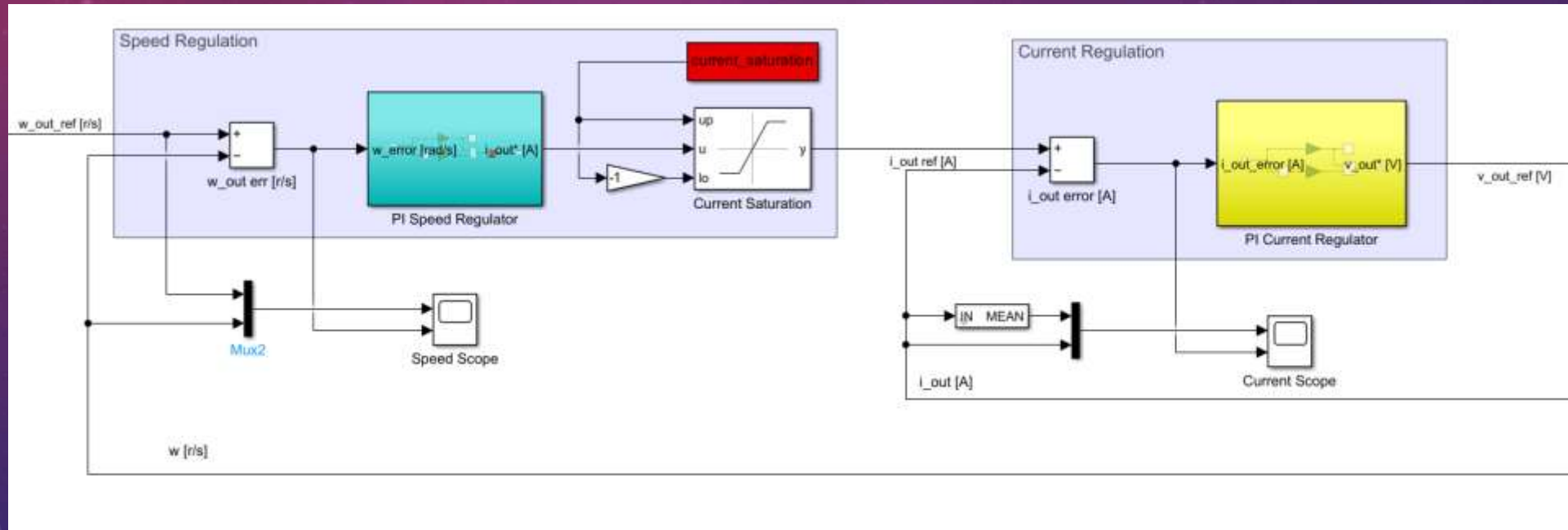
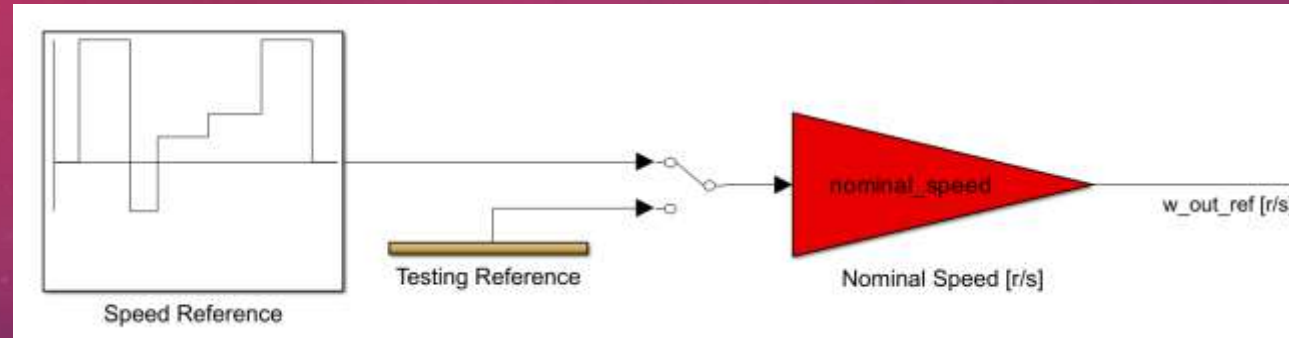
TOTAL SIMULATION SCHEME

(global views, zooms are in the following pages)



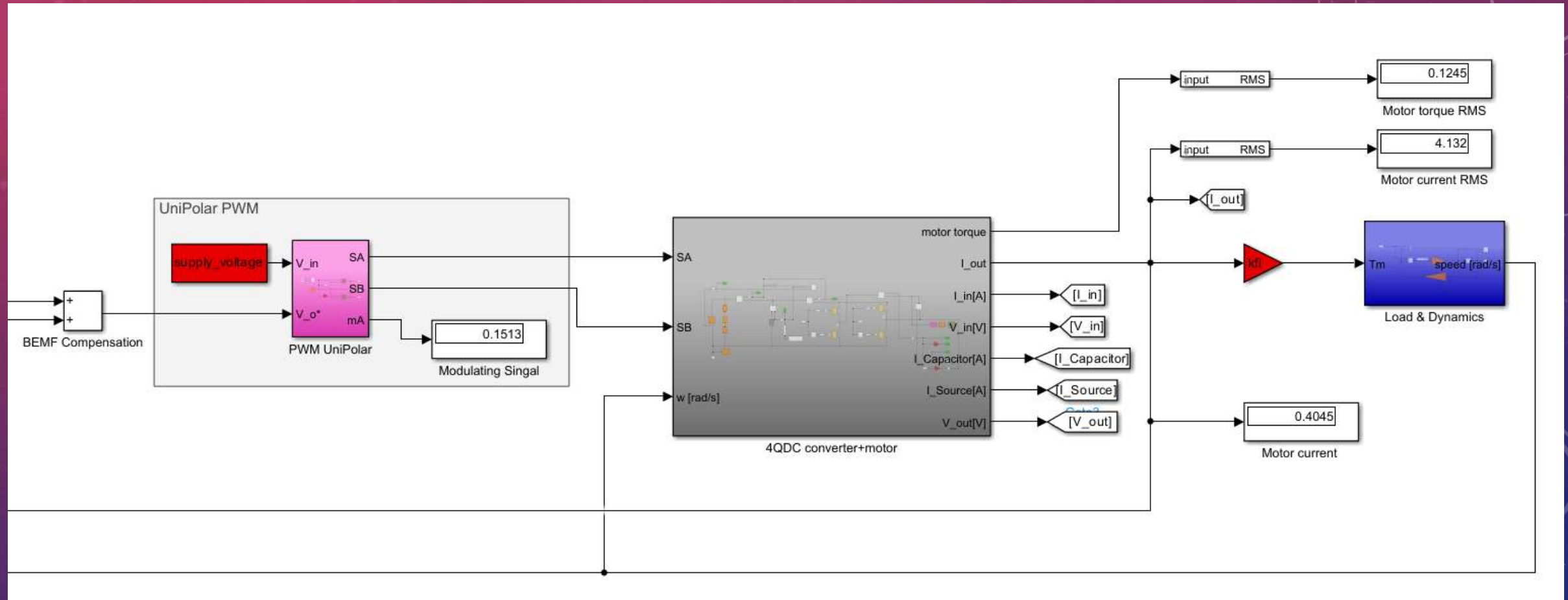
TOTAL SIMULATION SCHEME

(global views and zooms)



TOTAL SIMULATION SCHEME

(global views and zooms)



The display with the RMS blocks on the top right are used to calculate the RMS values of motor current and torque.

SIMULATION PARAMETERS

Physical Parameters

```
% MOTOR DATA %
nominal_speed = 6930*2*pi/60;           % [r/s] %
winding_resistance = 0.32;               % [Ohm] %
winding_inductance = 0.082 * 10^(-3);    % [Henry] %
rotor_inertia = 138;                     % [g·cm2] %
load_torque_slope = 5.856*10^(-5);       % [Nm/(rad/s)] ≡ [Nm·s] %
% kfi = 1/581;                           % [V/rpm] %
kfi = 1/317 * (60/(2*pi));               % [V/(rad/s)] ≡ [Nm/A] %

% SUPPLY SYSTEM %
supply_voltage = 30;                     % [Volt] %
source_internal_resistance = 0.30;        % [Ohm] %
source_internal_inductance = 0.30 * 10^(-3); % [Henry] %
filter_capacitance = 1e-2;               % [Farad] %
filter_capacitance_serie_resistance = 0.001; % [Ohm] %

% 4Q CONVERTER %
current_saturation = 9.2;                 % [A] %

% LOAD DATA %
load_inertia = 200;                       % [g·cm2] %
total_inertia = (rotor_inertia + load_inertia) * 10^(-7); % [kg·m2] %
```

Regulation Parameters

```
s = tf('s');

% CARRIER & SAMPLING TIME %
carrier_frequency = 400 * 10^3;           % [Hertz] %
carrier_period = 1/carrier_frequency;     % [seconds] %
sampling_time = carrier_period / 50;      % [seconds] %

% CURRENT REGULATION %
Gc = 1/(winding_inductance*s + winding_resistance); % Motor Eq Circuit Transfer Function. %
ki_c = 3050;                              % PI Current Regulator ki constant %
z_c = -54900;                             % PI Current Regulator zero. %
kp_c = 18;                                % PI Current Regulator kp constant %

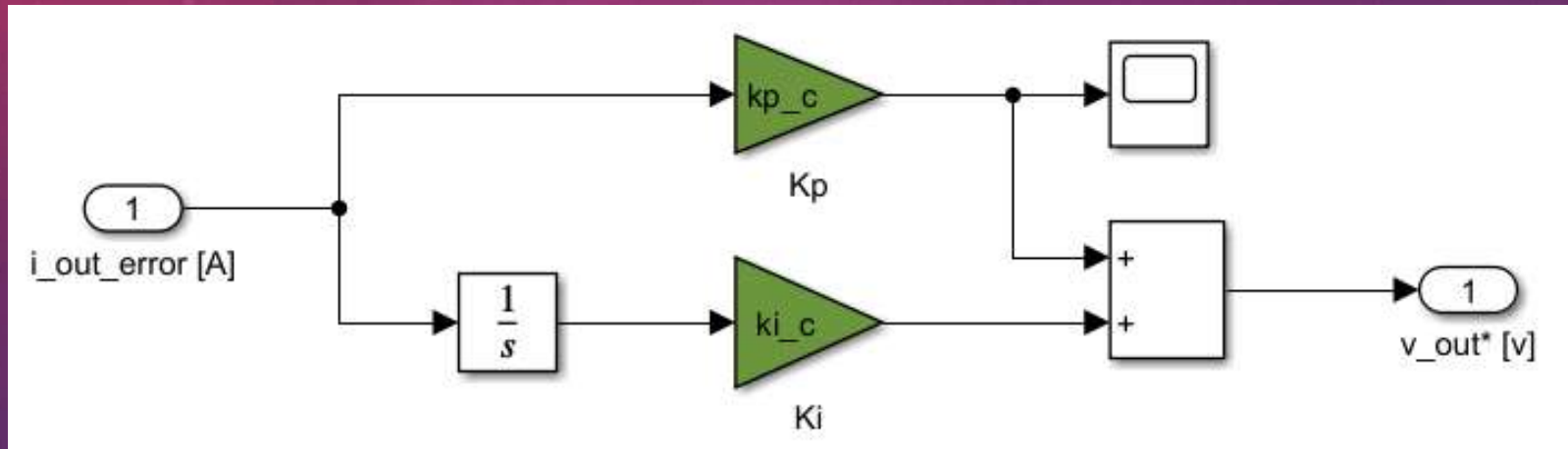
% SPEED REGULATION %
Gw = 1/(total_inertia*s + load_torque_slope); % Load Eq Circuit Transfer Function. %
ki_w = 0.0005;                             % PI Speed Regulator ki constant. %
z_w = -9*10^(-5);                          % PI Speed Regulator zero %
kp_w = 0.180;                              % PI Speed Regulator kp constant. %
```



PI CURRENT REGULATOR SCHEME

(inner loop regulator)

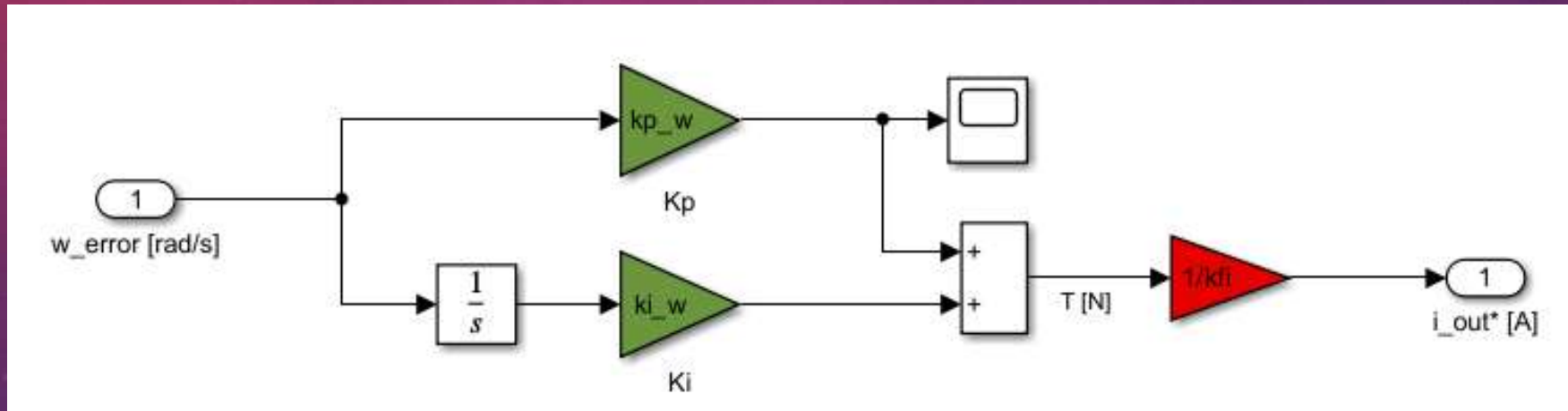
$$K_{ic} = 3050 \quad K_{pc} = 18$$



PI SPEED REGULATOR SCHEME

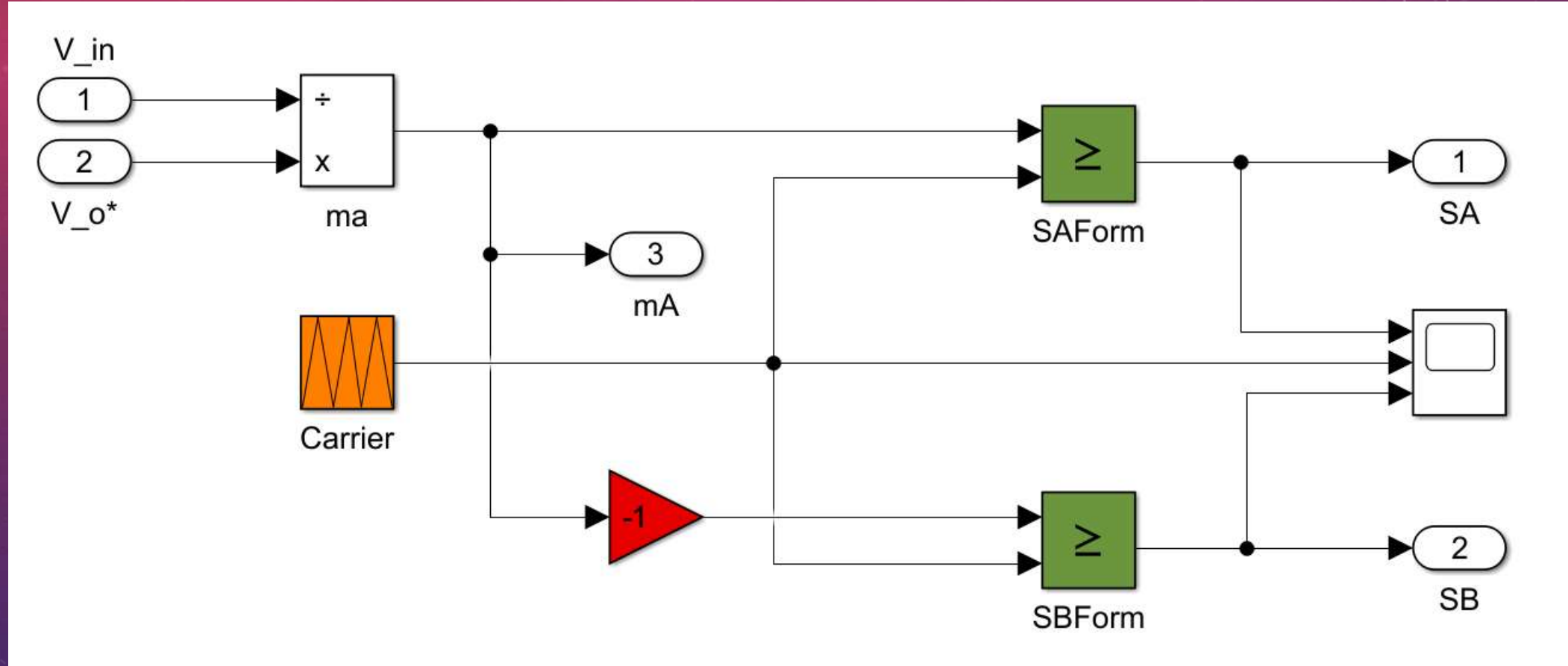
(outer loop regulator)

$$K_{iw} = 0,0005 \quad K_{pw} = 0,150$$



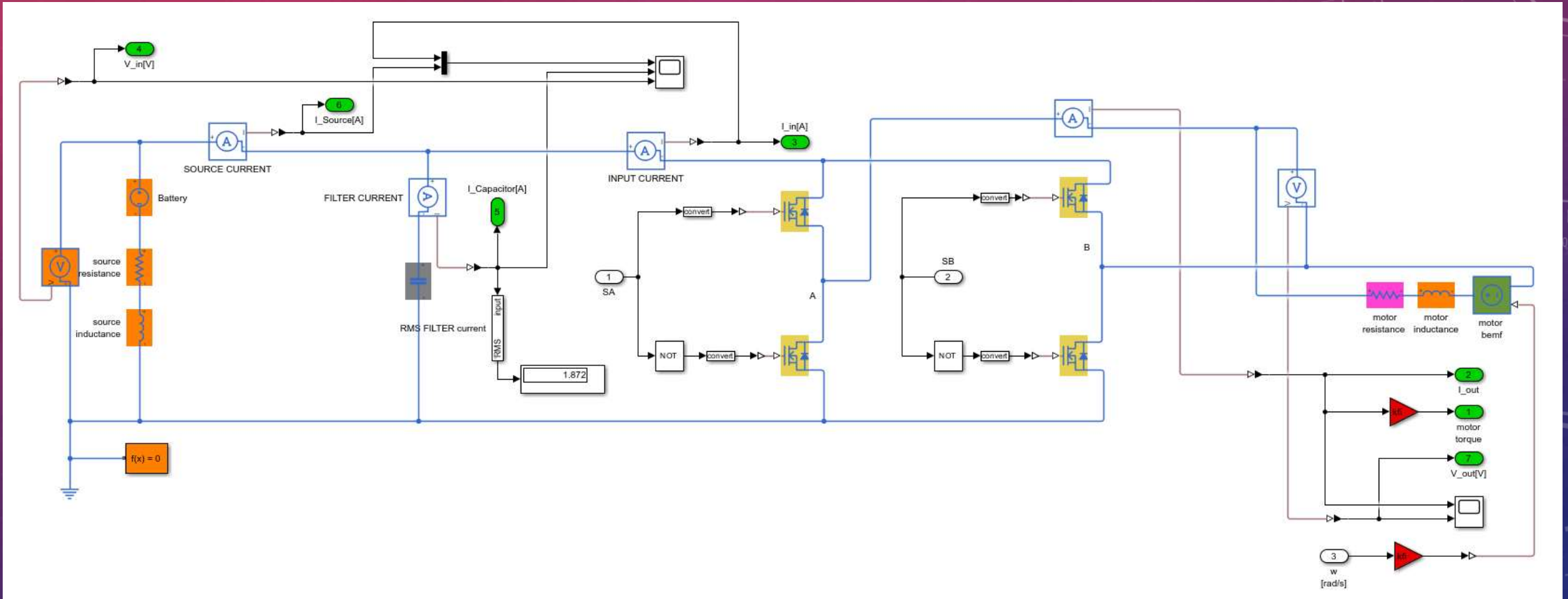
PWM UNI-POLAR

(with a triangular carrier)



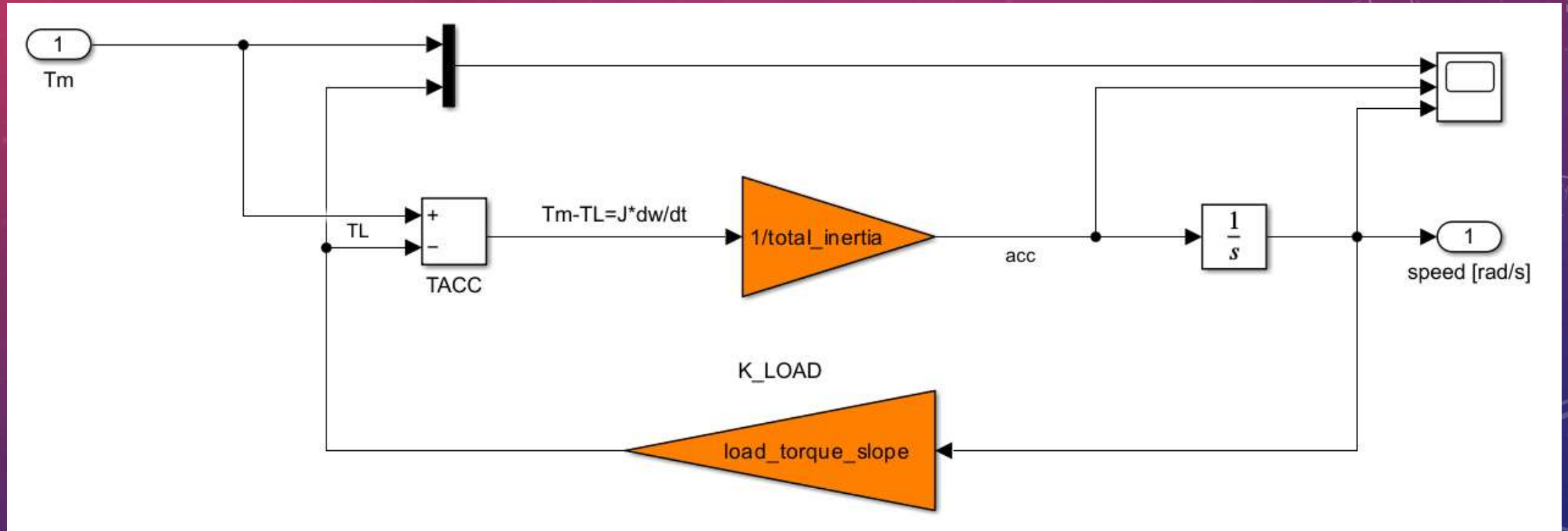
SUPPLY SYSTEM, 4Q CONVERTER & MOTOR

(all the simscape part of the scheme is contained in a single simulink block)



LOAD & DYNAMICS

(from torque to velocity calculation)



SPEED PLOT

(specifics 5.3 5.5 5.6 5.7)

(Each block of the x-axis is equal to 0,2 seconds)



SPEED PLOT

(specifics 5.3 5.5 5.6 5.7)

(Each block is equal to 0,2 seconds)



The Rise Rime is $\leq 100\text{ms}$ only for smaller step, even if the 4QCh is providing the maximum current, which is insufficient, as shown in the following slides.

SPEED PLOT

(specifics 5.3 5.5 5.6 5.7)



Speed Overshoot and Steady State speed Error are always lower than 1%

OUTPUT CURRENT PLOT

(specifics 5.1 5.2 5.4)

(Each block of the x-axis is equal to 0,2 seconds)



Note that if a lower sampling time is used, the current trend becomes much less irregular, but also simulation time significantly increases.

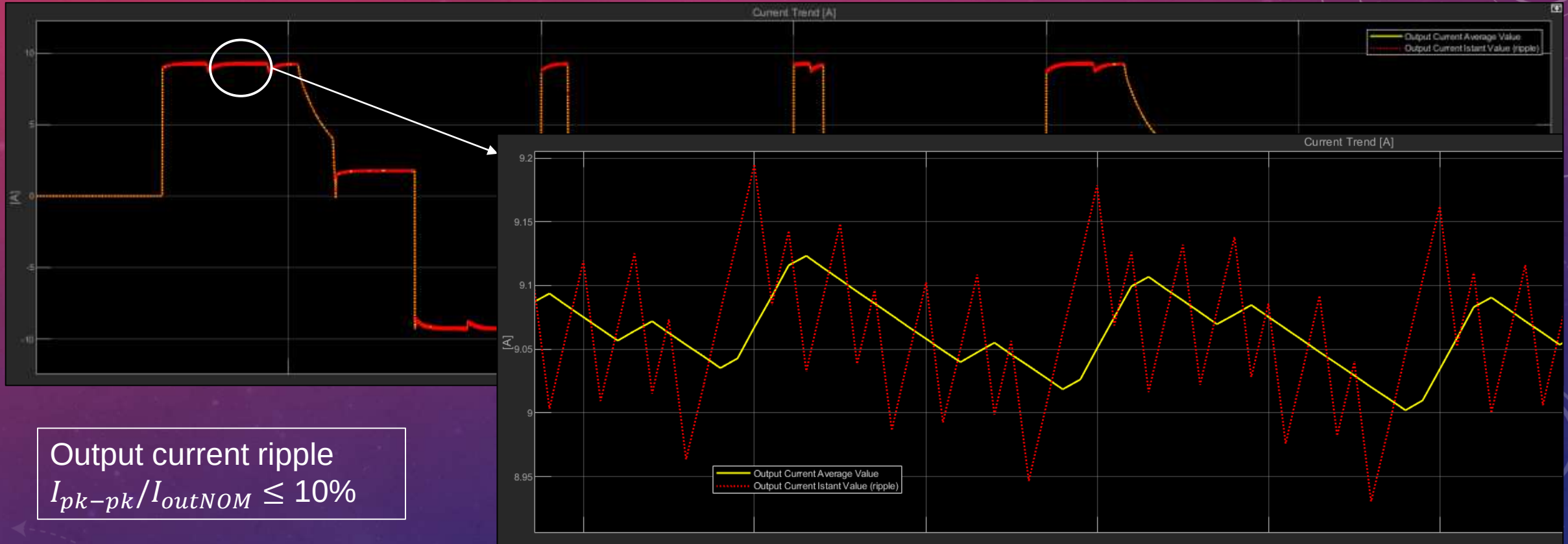
RMS motor current = 5,453 A
RMS motor torque = 0,1643 A

The RMS requested values were calculated on matlab, you can see the used blocks in slide 4.



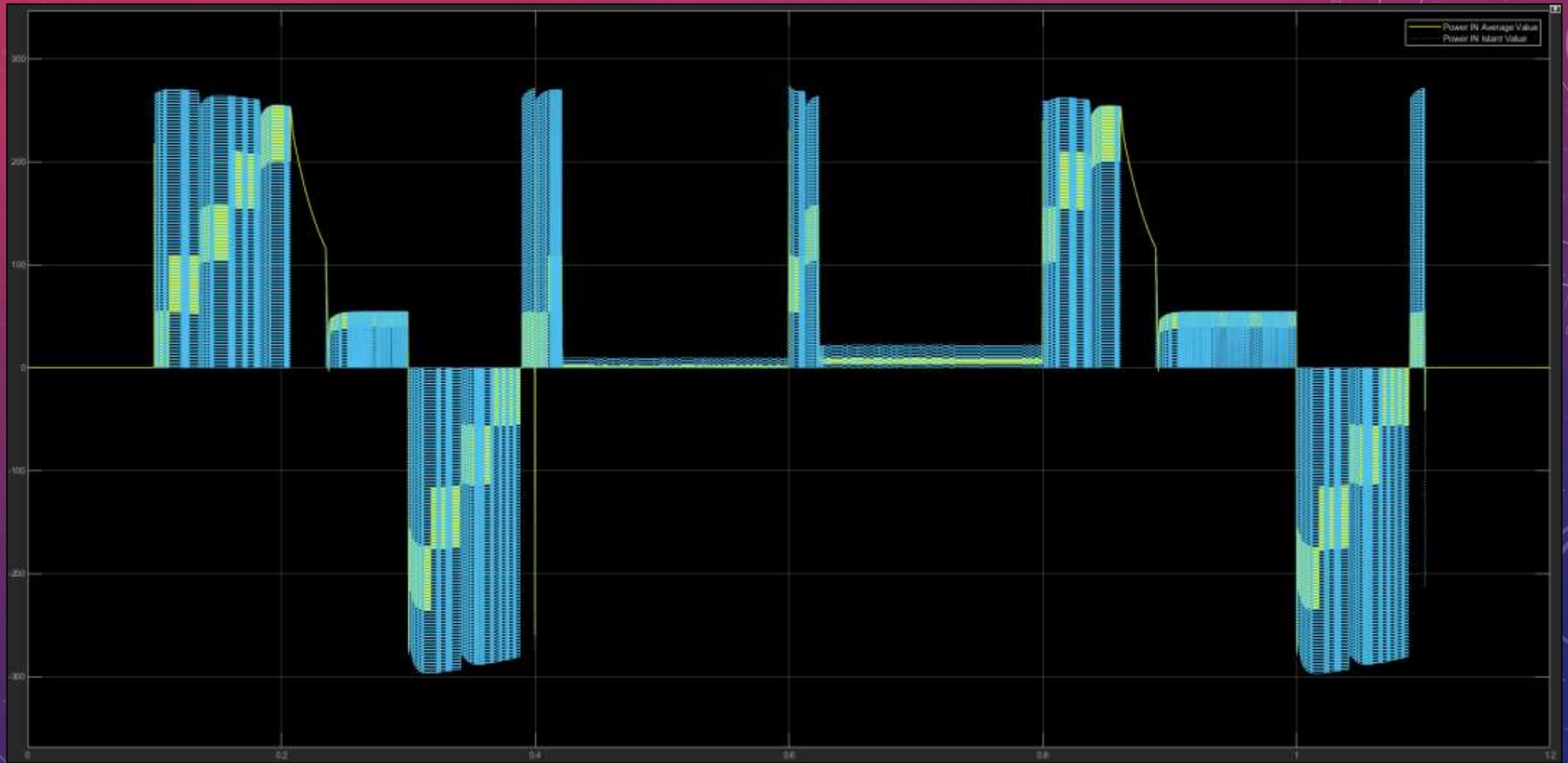
OUTPUT CURRENT PLOT

(specifics 5.1 5.2 5.4)

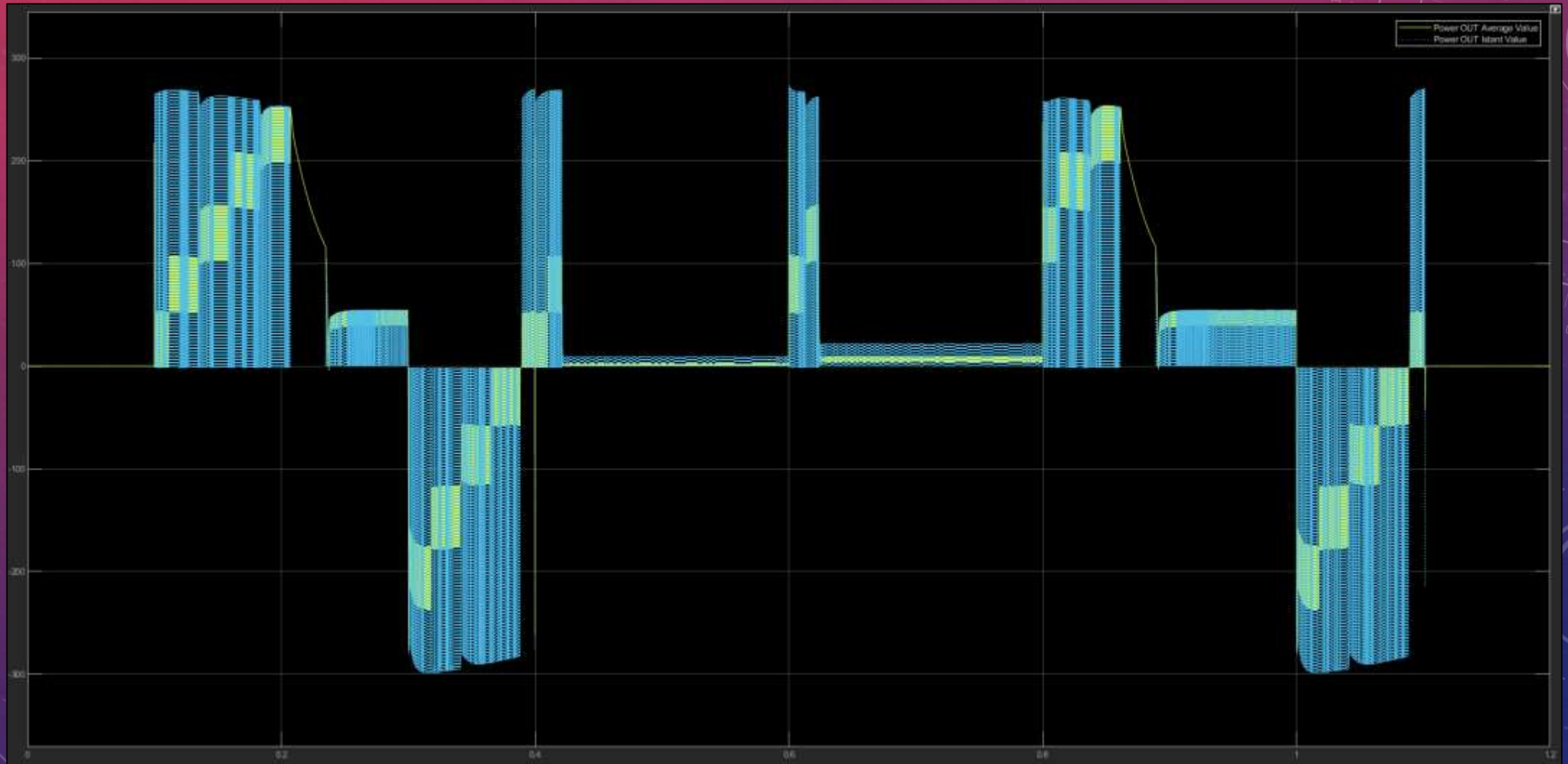


Output current ripple
 $I_{pk-pk}/I_{outNOM} \leq 10\%$

IN-POWER & OUT-POWER PLOT (W)



IN-POWER & OUT-POWER PLOT (W)





ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA